# Input-gradient space particle inference for neural network ensembles

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## Overview

**TL;DR**: We learn an ensemble of neural networks that is **diverse** with respect to their **input gradients**.

## Repulsive deep ensembles (RDEs) [1]

**Description**: Train an ensemble $\{\boldsymbol{\theta}_i\}_{i=1}^M$ using Wasserstein gradient descent [2], which employs a kernelized repulsion term to diversify the particles to cover the Bayes posterior $p(\boldsymbol{\theta}|\mathcal{D})$

$$\boldsymbol{\theta}_i^{(t+1)} = \boldsymbol{\theta}_i^{(t)} + \eta_t \left( \underbrace{\nabla_{\boldsymbol{\theta}_i^{(t)}} \log p(\boldsymbol{\theta}_i^{(t)} \mid \mathcal{D})}_{\text{Driving force}} - \underbrace{\frac{\sum_{j=1}^N \nabla_{\boldsymbol{\theta}_i^{(t)}} k(\boldsymbol{\theta}_i^{(t)}, \boldsymbol{\theta}_j^{(t)})}{\sum_{j=1}^N k(\boldsymbol{\theta}_i^{(t)}, \boldsymbol{\theta}_j^{(t)})}}_{\text{Repulsion force}} \right)$$

- The driving force directs the particles towards high density regions of the posterior.
- The repulsion force pushes the particles away from each other to enforce diversity.

**Problem**: It is unclear how to define the repulsion term for neural networks:
- weight-space repulsion is ineffective due to overparameterization.
- function-space repulsion often results in underfitting.

## Defining the input-gradient kernel $k$

Given a base kernel $\kappa$, we define the kernel in the input-gradient space for a minibatch of training samples $\mathcal{B} = \{(\mathbf{x}_b, y_b)\}_{b=1}^B$ as follows:

$$k(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) = \boxed{\frac{1}{B}\sum_{b=1}^{B}} \boxed{\kappa(\nabla_{\mathbf{x}_b}\mathbf{f}(\mathbf{x}_b;\boldsymbol{\theta}_i)_{y_b}, \nabla_{\mathbf{x}_b}\mathbf{f}(\mathbf{x}_b;\boldsymbol{\theta}_j)_{y_b})}$$

Take the average over all samples in the mini-batch.

Compare the input gradients of two particles with respect to the same input.

We choose the RBF kernel on a unit sphere as the base kernel $\kappa$:

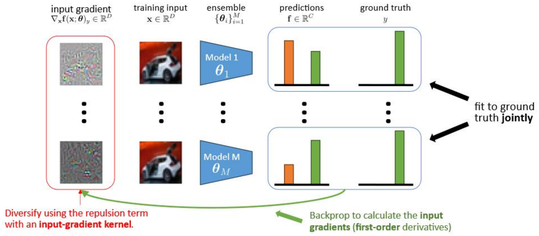
$$\kappa(\mathbf{s}_i, \mathbf{s}_j; \boldsymbol{\Sigma}) = \boxed{\exp\left(-\frac{1}{\boxed{2h}}(\mathbf{s}_i - \mathbf{s}_j)^\top \boxed{\boldsymbol{\Sigma}}^{-1}(\mathbf{s}_i - \mathbf{s}_j)\right)}, \quad \boxed{\mathbf{s}_i = \frac{\nabla_{\mathbf{x}}\mathbf{f}(\mathbf{x};\boldsymbol{\theta}_i)_y}{||\nabla_{\mathbf{x}}\mathbf{f}(\mathbf{x};\boldsymbol{\theta}_i)_y||_2}}$$

A scalar adaptively adjusted to prevent kernel vanishing.

Diagonal matrix containing the lengthscales.

Normalize input gradients to unit vectors.

## First-order Repulsive deep ensembles (FoRDEs)




**Possible advantages:**
- Each member is guaranteed to represent a different function;
- The issues of weight- and function-space repulsion are avoided;
- Each member is encouraged to learn different features, which can improve robustness.

## Main takeaways

1. Input-gradient-space repulsion can perform better than weight- and function-space repulsion.
2. Better corruption robustness can be achieved by configuring the repulsion kernel using the eigen-decomposition of the training data.

## Benchmark comparison

Table 1: **FoRDE-PCA achieves the best performance under corruptions while FoRDE-Identity outperforms baselines on clean data. FoRDE-Tuned outperforms baselines on both clean and corrupted data.** Results of RESNET18 / CIFAR-100 averaged over 5 seeds. Each ensemble has 10 members. cA, cNLL and cECE are accuracy, NLL, and ECE on CIFAR-100-C.

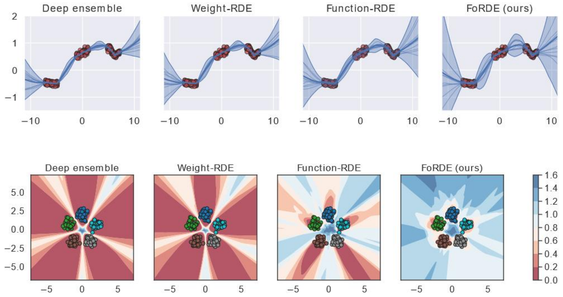
| METHOD | NLL ↓ | ACCURACY (%) ↑ | ECE ↓ | cA / cNLL / cECE |
|---|---|---|---|---|
| DEEP ENSEMBLES | **0.70**±0.00 | 81.8±0.2 | 0.041±0.003 | 54.3 / 1.99 / 0.05 |
| WEIGHT-RDE | **0.70**±0.01 | 81.7±0.3 | 0.043±0.004 | 54.2 / 2.01 / 0.06 |
| FUNCTION-RDE | 0.76±0.02 | 80.1±0.4 | 0.042±0.005 | 51.9 / 2.08 / 0.07 |
| FORDE-PCA (OURS) | 0.71±0.00 | 81.4±0.2 | 0.039±0.002 | **56.1 / 1.90 / 0.05** |
| FORDE-IDENTITY (OURS) | **0.70**±0.00 | **82.1**±0.2 | 0.043±0.001 | 54.1 / 2.02 / 0.05 |
| FORDE-TUNED (OURS) | **0.70**±0.00 | **82.1**±0.2 | 0.044±0.002 | 55.3 / 1.94 / 0.05 |

Table 2: **FoRDE-PCA achieves the best performance under corruptions while FoRDE-Identity has the best NLL on clean data. FoRDE-Tuned outperforms most baselines on both clean and corrupted data.** Results of RESNET18 / CIFAR-10 averaged over 5 seeds. Each ensemble has 10 members. cA, cNLL and cECE are accuracy, NLL, and ECE on CIFAR-10-C.

| METHOD | NLL ↓ | ACCURACY (%) ↑ | ECE ↓ | cA / cNLL / cECE |
|---|---|---|---|---|
| DEEP ENSEMBLES | 0.117±0.001 | 96.3±0.1 | 0.005±0.001 | 78.1 / 0.78 / 0.08 |
| WEIGHT-RDE | 0.117±0.002 | 96.2±0.1 | 0.005±0.001 | 78.0 / 0.78 / 0.08 |
| FUNCTION-RDE | 0.128±0.001 | 95.8±0.2 | 0.006±0.001 | 77.1 / 0.81 / 0.08 |
| FEATURE-RDE | 0.116±0.001 | 96.4±0.1 | **0.004**±0.001 | 78.1 / 0.77 / 0.08 |
| FORDE-PCA (OURS) | 0.125±0.001 | 96.1±0.1 | 0.006±0.001 | **80.5 / 0.71 / 0.07** |
| FORDE-IDENTITY (OURS) | **0.113**±0.002 | 96.3±0.1 | 0.005±0.001 | 78.0 / 0.80 / 0.08 |
| FORDE-TUNED (OURS) | 0.114±0.002 | 96.4±0.1 | 0.005±0.001 | 79.1 / 0.74 / 0.07 |

References
[1] F. D'Angelo and V. Fortuin, "Repulsive deep ensembles are Bayesian," Advances in Neural Information Processing Systems, vol. 34, pp. 3451–3465, 2021.
[2] C. Liu, J. Zhuo, P. Cheng, R. Zhang, and J. Zhu, "Understanding and Accelerating Particle-Based Variational Inference," in International Conference on Machine Learning, 2019.

## Illustrative experiments




For a 1D regression task (above) and a 2D classification task (below), FoRDEs capture higher uncertainty than baselines in all regions outside of the training data. For the 2D classification task, we visualize the entropy of the predictive posteriors.

## Tuning the lengthscales $\boldsymbol{\Sigma}$

Each lengthscale is inversely proportional to the strength of the repulsion force in the corresponding input dimension.

$$\underbrace{\frac{\partial}{\partial s_d}\kappa(\mathbf{s}, \mathbf{s}'; \boldsymbol{\Sigma})}_{\text{Repulsion force in the } d\text{-th dimension}} = -\frac{s_d - s'_d}{h\Sigma_{dd}}\kappa(\mathbf{s}, \mathbf{s}'; \boldsymbol{\Sigma}) \propto \frac{1}{\boxed{\Sigma_{dd}}}$$

lengthscale in the $d$–th dimension

**Proposition**: One should apply strong forces in high-variance dimensions (more in-between uncertainty) and weak forces in low-variance dimensions (less in-between uncertainty).

In high-variance data dimensions, distances between data points are large, which lead to more in-between uncertainty → we can apply strong repulsion force to push the input gradients far away from each other.

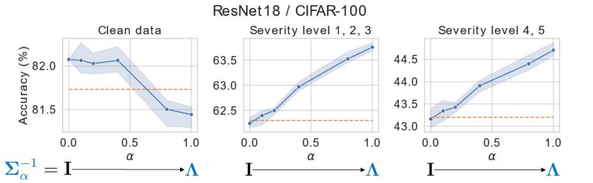
In low-variance data dimension, data points lie close to each other, leading to less in-between uncertainty → we need to use weak repulsion force.

- Use PCA to get the eigenvalues and eigenvectors of the training data: $\{\mathbf{u}_d, \lambda_d\}_{d=1}^D$
- Define the base kernel:

$$\kappa_{\text{PCA}}(\mathbf{s}, \mathbf{s}'; \boldsymbol{\Sigma}_\alpha) = \exp\left(-\frac{1}{2h}(\mathbf{U}^\top\mathbf{s} - \mathbf{U}^\top\mathbf{s}')^\top \boldsymbol{\Sigma}_\alpha^{-1}(\mathbf{U}^\top\mathbf{s} - \mathbf{U}^\top\mathbf{s}')\right)$$

- $\mathbf{U} = [\mathbf{u}_1 \ \ \mathbf{u}_2 \ \ \cdots \ \ \mathbf{u}_D]$ is a matrix containing the eigenvectors as columns.
- $\boldsymbol{\Sigma}_\alpha^{-1} = (1-\alpha)\mathbf{I} + \alpha\boldsymbol{\Lambda}$ where $\boldsymbol{\Lambda}$ is a diagonal matrix containing the eigenvalues.

## Lengthscale tuning experiments




- Blue lines show accuracies of FoRDEs, while dotted orange lines show accuracies of Deep ensembles.
- When moving from the identity lengthscale $\mathbf{I}$ to the PCA lengthscales $\boldsymbol{\Lambda}$:
  - FoRDEs exhibit small performance degradations on clean images of CIFAR-100;
  - while becomes more robust against the natural corruptions of CIFAR-100-C.